Cognita — Governance Document

AI Systems for Reviewers · Reasoning-audit prototype · v0.1 (October 2026)

1. Purpose and Scope

Cognita helps educators and research reviewers evaluate **how** a person reasoned rather than whether the final answer is correct. It maps the reasoning chain (claims, evidence, inferences, assumptions, conclusions), scores six cognitive dimensions with quoted evidence, surfaces gaps, and generates viva questions. It is an **advisory** tool: no output is final until a human reviewer signs off.

Out of scope: automated grading, plagiarism or AI-authorship detection, admission or employment decisions, and any profiling of individuals.

2. Prompts

A single fixed system prompt is used server-side; users cannot modify it. The model returns strict JSON validated before display. Model: openai/gpt-6-astra via the Lovable AI Gateway, low reasoning effort.

```
You are Cognita, a reasoning auditor that assists human reviewers.
You DO NOT grade the final answer or give a pass/fail. You audit HOW the author thought.
- Evaluate only the text provided. Never infer identity, gender, ethnicity,
  nationality, or native language.
- Do not penalize grammar, dialect, or writing style; judge reasoning substance only.
- Every score must cite a verbatim quote from the text as evidence.
- Be calibrated: state uncertainty. The human reviewer makes the final judgement.
Return ONLY JSON: summary, dimensions[6], reasoning_chain[], gaps[], questions_for_author[],
confidence
```

Rubric dimensions (0–5): Problem Framing, Evidence Use, Synthesis, Counter-argument, Methodology, Metacognition. Submission text is wrapped in triple-quote delimiters and treated as data, reducing prompt-injection risk (instructions inside a student essay are not followed).

3. Guardrails

Risk	Guardrail
Hallucinated judgement	Every score must quote verbatim evidence shown to reviewer for verification
Over-confidence	Model reports calibrated confidence; shown beside the score
Prompt injection	Fixed server prompt; submission delimited as data; JSON-only output
Misuse as auto-grader	No pass/fail or grade produced; sign-off required
Abuse / cost	Input length 80–12,000 chars validated server-side; 429/402 handled gracefully
Key exposure	API key held only on server; never shipped to browser

4. Human-in-the-Loop Controls

• **Advisory by design:** the interface labels all AI output as advisory and shows the evidence behind each score.

• **Score override:** each dimension has a slider; overridden AI scores remain visible (struck through) so the deviation is transparent and auditable.

• **Mandatory rationale:** the 'Approve & sign' action is disabled until the reviewer writes their own judgement (min. 10 chars).

• **Dialogue over verdict:** 'Viva questions' push reviewers to talk with the author before concluding, giving the student a voice.

• **Contestability:** institutions should let authors see the signed audit and appeal; the evidence quotes make disputes concrete.

• **Escalation:** low confidence (<60%) or many 'unsupported' steps should trigger a second human reviewer (recommended policy).

5. PII and Privacy

• **Redaction before AI:** emails, phone numbers and student/matriculation IDs are replaced server-side with [EMAIL], [PHONE], [ID] tokens before the text leaves the server; the count is shown to the reviewer.

• **No storage:** the prototype has no database. Submissions and results exist only in the reviewer's browser session and are discarded on reload.

• **No accounts, no tracking** of authors. Reviewers are advised to strip names before pasting.

• **Data minimisation:** only text and optional assignment context are sent; no metadata.

• **Transport:** HTTPS only; the gateway is used for inference with no training on submitted data.

• **Compliance alignment:** designed toward GDPR principles (purpose limitation, minimisation, storage limitation) and FERPA-style education records protection. Production deployment would add a DPA, retention policy and DPIA.

6. Bias Mitigation

• **Style-blind instruction:** the model is told not to penalise grammar, dialect or non-native phrasing, which disproportionately disadvantage multilingual and neurodivergent writers.

• **No identity inference:** the prompt forbids inferring demographics; PII redaction removes many identity cues.

• **Evidence anchoring:** requiring verbatim quotes limits impressionistic scoring and makes biased reasoning visible to the reviewer.

• **Process over product:** auditing reasoning rewards visible thinking (hedging, self-correction, uncertainty) instead of polished prose.

• **Planned evaluation:** paired-text testing (same reasoning, different dialect/fluency) to measure score drift; target absolute score difference < 0.5 per dimension; inter-rater agreement (Cohen's kappa) between AI and expert reviewers on a calibration set.

• **Monitoring:** override rates by dimension are a bias signal; frequent downward overrides on one dimension trigger rubric review.

7. Limitations and Accountability

LLMs can misread discipline-specific reasoning, miss implicit steps, and vary between runs. Scores are not psychometrically validated. Cognita must not be the sole basis for any high-stakes decision. Accountability rests with the signing reviewer and the deploying institution; this document is reviewed each release.